Theorem 1.3: If f and g are real-valued functions, then $f(g+h) = fg + fh$.

Proof. Let X be any set and f and g be real-valued functions defined on X. Let $x \in X$.

$$\begin{aligned}
(f(g+h))(x) &= f(x)(g+h)(x) && \text{(by the definition of product)}\\
&= f(x)[g(x)+h(x)] && \text{(by the definition of addition)}\\
&= f(x)g(x) + f(x)h(x) && \text{(by the distributivity of reals)}\\
&= (fg)(x) + (fh)(x) && \text{(by the definition of product)}\\
&= (fg+fh)(x) && \text{(by the definition of addition)}
\end{aligned}$$

Thus $(f(g+h))(x) = (fg+fh)(x)$ for all $x \in X$; hence $f(g+h) = fg + fh$. ∎

1.6.7 Some Special Functions

Now let us see some special functions.

(i) The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n,$$

where a_i are constants, is called a ***polynomial function***. Since the right hand side of the equality defining the function is a polynomial, this function is called a polynomial function.

(ii) The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = ax + b$ where $a \neq 0$ and b are constants, is called a ***linear function***. A function which is not linear is called a ***non-linear function***.

Clearly a linear function is a polynomial function. The graph of this function is a straight line; a straight line is called a linear curve; so this function is called a linear function. (one may come across different definitions for linear functions in higher study of mathematics.)

(iii) Let a be a non-negative constant. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = a^x$. If $a = 0, x \neq 0$ then the function becomes the zero function and if $a = 1$, then function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = a^x$ is the constant function $f(x) = 1$. [See, Figures 1.42 and 1.43].

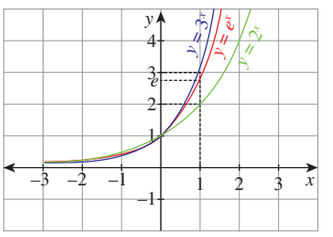


Figure 1.42

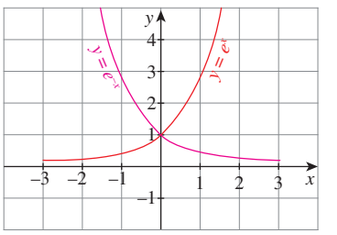


Figure 1.43

When $a > 1$, the function $f(x) = a^x$ is called an ***exponential function***. Moreover, any function having x in the "power" is called as an exponential function.



e is a special irrational number lies between 2 and 3. We will study more about e in the subsequent chapters.

(iv) Let $a > 1$ be a constant. The function $f : (0, \infty) \to \mathbb{R}$ defined by $f(x) = \log_a x$ is called a ***logarithmic function***. In fact, the inverse of an exponential function $f(x) = a^x$ on a suitable domain is called a logarithmic function. [See, Figure 1.44].

(v) The real valued function f defined by $f(x) = \frac{p(x)}{q(x)}$ on a suitable domain, where $p(x)$ and $q(x)$ are polynomials, $q(x) \neq 0$, is called a ***rational function***. In fact, the domain of this function is the set obtained from $\mathbb{R}$ by removing the real numbers at which $q(x) = 0$.